

Methodology

For our project we will be using Agile Development so we can have a flexible and adaptive approach. The development process will be divided into sprints.

The agile approach to software development consists of making incremental, small changes throughout the production of a project to keep development moving at a consistent pace throughout.

We will have X number of sprints which are checkpoints of sorts throughout the development phases. At the end of each sprint, we will ask various people of different demographics (if allowed) to review our project so far. This will allow us to know what we done good & what we can do better, getting viewpoints from various ages & knowledge on using web browsers.

Team Roles

- Samuel Wright: database & backend developer / assist in frontend / assist in documentation & design.
- Milosz Ziolkowski: database & backend / documentation & design.
- Omi Hossain: Assist in Frontend / UI Design and Documentation
- Team roles up for change / interpretation

Restaurant Booking System Sprints:

Sprint 1 (Weeks 1-4) Project planning and Research:

Goal: - For this sprint the goal here is to lay a strong foundation for the project

Tasks:

- Conduct detailed research on restaurant management systems.
- Meet as a group to clearly outline the project's scope and determine the most important features to develop first.
- Finalise and decide on the Tech Stack (React, HTML, CSS, JavaScript, PHP, Laravel, MySQL)
- Set up the Git repository
- Outlining the step-by-step plan for how your project will be completed over time and deciding who will be responsible for each part of the project.

Sprint 2 (Weeks 5-8): Front End UI Design:

Goal: - Design mock-ups and the user interface for the website.

Tasks:

- Create detailed UI Designs for the website (e.g: Home page, food menu page, Booking page)
- Review my designs with my group and get feedback and suggestions on how I can improve the user interface.
- Begin building the front-end in HTML, CSS, JavaScript and react.

Deliverables:

- Initial front-end structure with navigation.

Sprint 3 – (Weeks 9-12): Back – End Setup & Database Design:

Goal: - Set up the back-end infrastructure and design the database.

Tasks:

- Install and configure/ set up Laravel and MySQL.
- Design the database schema (e.g: table for users, reservations, menu, admin)
- Set up the authentication system (user sign-up, login, roles for customers, staff, and admin).
- Create basic API endpoints to retrieve the data (e.g: for the menu and for reservations)

Deliverable:

- Database schema and back – end setup.
- User authentication and API structure.

Sprint 4 (Weeks 13-16): Menu Management & Inventory System.

Goal: - Implement the main features for menu management and inventory control.

Tasks:

- Build the menu management system where admin can add/edit/remove items.
- Link the menu data to the **MySQL database**.
- Implement the **inventory control** to track stock levels of menu items.
- Design and implement the **menu display page** for customers.

Deliverables:

- Menu management system (admin side).

- Customer-facing menu display lined to inventory.

Sprint 5 (Weeks 17–20): Reservation System.

Goal: Develop and implement the table booking system.

Tasks:

- Create a reservation system where customers can book tables
- Implement a form validation to ensure valid bookings
- Store reservation data in the database and display available time slots.
- Build an admin dashboard for viewing and managing reservations.

Deliverables:

- Fully functioning reservation system.
- Admin dashboard for reservations.

Sprint 6 (Weeks 21-24): Staff Management & User Roles:

Goal: - Develop a system for managing staff and defining roles.

Tasks:

- Create the staff management module where admin can manage staff (such as: add/edit/remove).
- Define user roles (admin, staff, customer) and permissions for different sections of the system.
- Implement features to assign tasks and track staff activity (e.g., reservation handling, order processing).

Deliverables:

- Staff management system with role-based permissions.
- Tracking system for staff activities.

Sprint 7 (Weeks 25-30): Transactions & Billing

Goal: - Implement the payment system and generate bills.

Tasks:

- Develop the billing system that generates bills for customers after orders/reservations

- Link the transaction data to the database for tracking payments.
- Implement an interface for customers to **view their bills** and track order history.

Deliverables:

- Transaction and billing system.
- Bill history available for customers and admin.

Sprint 8 (Weeks 31-36): Testing & Debugging

Goal: - Test all features and fix any bugs.

Tasks:

- Conduct unit tests for each system (reservations, menu, billing etc...)
- Perform integration tests to ensure all components work together.
- Debug any issues and ensure that the system is stable.
- Organize user acceptance testing (UAT) where group members or external users can test the system.

Deliverables:

- Tested and debugged system, ready for deployment.

Sprint 9 (Weeks 37-40): Deployment & Final Touches

Goal: - Deploy the system and make final adjustments.

Tasks:

- Host the website (e.g.: GitHub)
- Perform final testing for the front-end performance and responsiveness.
- Write the final project documentation
- Conduct a final project review with the group to ensure everything is working as expected.

Deliverables:

- Deployed restaurant booking system.
- Final project documentation.

Sprint Review and Feedback

- At the end of each sprint, we will conduct a sprint review to discuss what was accomplished, what went well, and what challenges were faced.

We will use tools like **Trello** or **Jira** for sprint tracking and management, and **Gitlab Projects** to monitor the development process.